Food Preservation, Nutrition, Medicine, etc.

Notes

Food Preservation :

- Food preservation may be defined as the process of treating and handling food in such a way as to stop, control or greatly slow down spoilage and, of course, to minimize the possibility of foodborne illness, while maintaining the optimum nutritional value, texture and flavour.

Principles of Food Preservation :

- The principles of food preservation are as follows–

1. **Removal of microorganisms or inactivating them :** This is done by removing air, water (moisture), lowering or increasing temperature, increasing the concentration of salts, sugar or acids in food. For the preservation of green leafy vegetables, the water should be removed from the leaves so that the microorganisms cannot survive.
2. **Inactivating enzymes :** Enzymes found in food can be inactivated by changing its conditions such as temperature and moisture. One of the methods of preserving peas is to put them for a few minutes in boiling water. This method inactivates enzymes.

Methods of Food Preservation :

- **Pasteurization :** Pasteurization of milk requires temperatures of about 63°C (145°F) maintained for 30 minutes or alternatively heating to a higher temperature i.e. 72°C (162°F) for 15 seconds. This method kills the bacteria.
- **Freeze :** The colder a food is, the slower is its rate of deterioration. In cold storage refrigerated air is used for food preservation.
- **Use of strong concentrations :** Use of strong concentration of salts, inhibit the growth of bacteria.
- **Canning :** Canning is most widespread and effective means of long term food storage. In canning food is cooked under pressure to attain a temperature high (around 121°C) to destroy endospores.
- **Organic acids :** Organic acids used to preserve food include benzoic acid, sorbic acid and propionic acid. These acids are added as salts such as sodium benzoate, potassium sorbate, sodium propionate. These acids inhibit the growth of bacteria. Some other organic compounds as cinnamon and cloves contain the benzene derivative **eugenol**, a potent microbial agent.
- **Drying method :** Food drying is a method of food preservation in which food is dried (dehydrated or desiccated). It inhibits the growth of bacteria, yeasts and mold (fungi) through the removal of water.
- Most fruits & vegetables age less rapidly when the level of oxygen in the atmosphere surrounding them is reduced. This is because the reduced oxygen slows down the respiration and metabolic rate of the product and, therefore, slows down the natural aging process.

Food Preservatives :

- Food preservatives are chemical substances added to a product to destroy or inhibit multiplication of microorganisms such as bacteria and fungi.
- Some common food preservatives are as follows :

(i) **Sodium meta bisulphite ($Na_2S_2O_5$) –** It is used in preservation of jams, sauce and pickles.

(ii) **Sodium benzoate ($C_7H_5NaO_2$) –** It is widely used food preservative. It is used to preserve soft drinks and acidic foods.

(iii) **Sorbic acid ($C_6H_8O_2$) and its salts –** Sorbic acid or 2,4-hexadienoic acid, is a natural organic compound used as a preservative. It is a colourless solid that is slightly soluble in water and sublimes readily. Its sodium and potassium salts are used as preservative to inhibit the growth of bacteria and fungi in cheese, cooked foodstuffs, pickles and food products of flesh and fish.

(iv) **Epoxides -** Epoxide of ethylene (ethylene oxide-C_2H_4O) and epoxide of propylene (propylene oxide - CH_3CHCH_2O) are good food preservatives to preserve food stuffs with little amount of water as dry fruits and spices.

Drugs :

- Drugs are natural or synthetic substances, when taken into a living body, affects its functioning or structure, and are used in the diagnosis, mitigation, treatment or prevention of a disease or relief of discomfort.
- Drugs are also called **medicines**.
- In medical science drugs have been classified on the basis of their functions, which are as follows–

1. **Antipyretics :**

- Antipyretics are substances that reduce fever. Examples : Aspirin (acetylsalicylic acid), paracetamol, phenacetin.

2. **Analgesics :**

- The term Analgesics encompasses a class of drugs that are designed to relieve pain without causing the loss of consciousness. It has been divided into two groups –

(a) **Non-narcotic analgesics :** Aspirin (2- acetoxybenzoic acid) and Paracetamol (2- Acetamidophenol).

(b) **Narcotic analgesics :** Drugs which are administered in small quantity to relieve pain and promoting sleepness, are called narcotic analgesics.

- Its excessive dose is responsible for laxiness, coma and may be causing death.
- Examples of narcotics analgesics are opium products such as morphine, heroin and codeine. These are also called opiates because these are derived from opium. These are habit forming substances.

3. **Antiseptic :**

- Antiseptics are antimicrobial substances that are applied to living tissue/skin to reduce the possibility of infection, sepsis or putrefaction (the process of decay or rotting in body).
- Antiseptics are harmless to living tissues and are used on cuttings and wounds.
- Examples of antiseptics are mercurochrome, mercuric chloride, alcohol, iodine, hydrogen peroxide, boric acid, potassium permangnate, iodoform etc.
- Garlic has powerful antiseptic properties.

4. **Disinfectants :**

- Disinfectants are antimicrobial agents that are applied to the surface of non-living objects to destroy microorganisms that are living on the objects.
- These are harmful to living tissues and cannot be applied on skin.
- Examples of disinfectants are phenol, methylphenol, chlorine, bleaching powder, formaldehyde etc.

5. **Tranquillizer :**

- A medicinal drug taken to reduce tension or anxiety is known as tranquillizer.
- It acts on the central nervous system and is used to calm, decrease anxiety or help a person to sleep.
- Reserpine is a strong tranquillizer which is obtained from the plant Rauwolfia serpentina. It is used for the treatment of high blood pressure.
- Barbituric acid and its derivatives as seconal and luminal are some other tranquillizers.

6. **Antibiotics :**

- Antibiotics are such chemicals that inhibit the growth of microorganisms or destroy them.
- Antibiotics are generally obtained or formed from living cells.
- Most modern antibiotics are semisynthetic modifications of microorganisms, but some are made completely through chemical synthesis.
- The first antibiotic penicillin was discovered by Alexander Flemming (1929) from Penicillium notatium, a fungus.
- Antibiotics do not work on viruses.

7. **Anaesthetics :**

- A substance that induces insensitivity to pain or a temporary loss of sensation is known as anaesthetics.
- Its effect is **reversible.** It means that affected organ gains its normal position after being less amount of anaesthetics substance.
- Anaesthetics may be divided into two broad classes: 'general' anaesthetics, which result in a reversible loss of consciousness, and 'local' anaesthetics, which cause a reversible loss of sensation for a limited reason of the body without necessarily affecting consciousness.

Question Bank

1. In fruits and vegetables, wax emulsion is used for –

(a) Creating shine on fruits and vegetables
(b) Extension of storage life
(c) Enhancing the ripening process
(d) None of the above

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (b)

In fruits and vegetables, wax emulsion is used for extension of their storage life. Wax coating is used as a carrier for sprout inhibitors, growth regulators and preservatives of fruits and vegetables. The principle disadvantage of wax coating is the development of off flaver if not applied properly. Fruits and vegetables can be stored for 10 to 12 days by wax coating.

2. Which one of the following substances is used in the preservation of food stuff?

(a) Citric Acid (b) Potassium Chloride
(c) Sodium Benzoate (d) Sodium Chloride

44th B.P.S.C. (Pre) 2000

Ans. (c)

The chemical formula of Sodium Benzoate is $C_7H_5NaO_2$. It is widely used as a food preservative, with E number E211. It is the sodium salt of benzoic acid and exists in this form when dissolved in water and its melting point is 410°C.

3. Which one of the following is used in preservation?

(a) Sodium Chloride (b) Sodium Benzoate
(c) Sodium Tartrate (d) Sodium Acetate

U.P. U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (b)

See the explanation of above question.

4. For the preservation of fruit juice which of the following is used ?

(a) Acetic acid (b) Formic acid
(c) Sulphuric acid (d) Sodium Benzoate

U.P.P.S.C. (R.I.) 2014

Ans. (d)

See the explanation of above question.

5. Which one of the following chemical is used in food preservation.

(a) Sodium Chloride (b) Caustic Soda
(c) Sodium Benzoate (d) Sulfuric Acid

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (c)

See the explanation of above question.

6. **Which one of the following is used in food preservation?**
(a) Sodium Carbonate (b) Acetylene
(c) Benzoic Acid (d) Sodium Chloride
U.P.P.C.S. (Pre) 1996
U.P.P.C.S. (Pre) 1992

Ans. (c)

Benzoic acid ($C_7H_6O_2$) or C_6H_5COOH is the colourless crystalline solid and simple aromatic carboxylic acid. The name is derived from gas benzoin. Its salts (e.g. sodium benzoate) are used as food preservative.

7. **Potato chips are packed in plastic bags in the atmosphere of :**
(a) Nitrogen Atmosphere (b) Hydrogen Atmosphere
(c) Oxygen Atmosphere (d) Iodine Atmosphere
U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (a)

Potato chips are packed in plastic bags in the atmosphere of nitrogen to protect them from being oxidized. It gives the chips a longer shelf life because bacteria, molds, etc. need oxygen to thrive, and in the absence of oxygen they cannot grow. It also keeps away moisture and keeps the chips intact during transportation.

8. **Monosodium glutamate (MSG) in food is used as :**
(a) Colour enhancer (b) Flavour enhancer
(c) Preserver (d) Emulsifier
M.P. P.C.S. (Pre) 2020

Ans. (b) & (c)

Monosodium Glutamate is commonly known as Ajinomoto. It is the sodium salt of glutamic acid. MSG is found naturally in some foods including tomatoes and cheese in this glutamic acid form. Monosodium glutamate (MSG) is a flavor enhancer and preserver commonly added to Chinese food, canned vegetables, soups and processed meats. It is used to flavor and season foods and to preserve them as well. It can enhance the perception of savoriness while preserving palatability. MSG is used in cooking as a flavor enhancer with an umami taste that intensifies the meaty, savory flavor of food, as naturally occurring glutamate does in foods such as stews and meat soups. MSG was first prepared in 1908 from seaweed broth by Japanese biochemist Kikunae Ikeda.

9. **Fruits stored in a cold chamber exhibit longer storage life, because :**
(a) Exposure to sunlight is prevented
(b) Concentration of CO_2 in environment is increased
(c) Rate of respiration is decreased
(d) There is an increase in humidity
U.P. P.C.S. (Pre) 2021

Ans. (c)

The shelf life of fruits and vegetables can be increased by keeping them in cold storage. This results in a slower ripening process and fruits and vegetables remain fresh for a longer duration. Notably, when fruits are stored in a cold chamber, their rate of respiration is decreased. When fruit respires they release ethylene which helps in ripening. Ethylene is also known as the 'fruit-ripening hormone'. Every fruit has a different level of ethylene production. The rate of ethylene production decreases when fruits are kept in cold storage. Thus, the shelf life of fruits increases.

10. **Refrigeration helps in food preservation by –**
(a) Killing the germs
(b) Reducing the rate of biochemical reactions
(c) Destroying enzyme action
(d) Sealing the food with a layer of ice
U.P.P.C.S.(Pre) 2013
U.P.P.C.S (Pre) 2011

Ans. (b)

Refrigeration preserves foods by slowing down the growth and reproduction of microorganisms or we can say that by refrigeration we can reduce the rate of biochemical reactions.

11. **Which of the following is a common refrigerant used in the domestic refrigerator?**
(a) Neon (b) Oxygen
(c) Freon (d) None of the above
Jharkhand P.C.S. (Pre) 2013

Ans. (c)

Freon is a common refrigerant used in domestic refrigerator. It is the name of a registered patent for a commercial refrigerant manufactured by Dupont. Freon is mildly toxic but stable halocarbon.

12. **'Triclosan', considered harmful when exposed to high levels for a long time, is most likely present in which of the following?**
(a) Food preservatives
(b) Fruit-ripening substances
(c) Reused plastic containers
(d) Toiletries
I.A.S. (Pre) 2021

Ans. (d)

Triclosan is a chemical with antibacterial properties. Triclosan is an ingredient added to many consumer products intended to reduce or prevent bacterial contamination. It is added to some antibacterial soaps and body washes, toothpastes, and some cosmetics products. It also can be found in clothing, kitchenware, furniture, and toys. In 2017, the US Food and Drug Administration (FDA) declared that triclosan is not generally recognized as safe and effective for antiseptic products intended for use in health care settings. In 2016, the

FDA also banned over-the-counter consumer antiseptic wash products containing triclosan from being marketed to consumers. These products include liquid, foam and gel hand soaps, bar soaps, and body washes. The basis of the ban was that manufacturers haven't proved that triclosan is safe for daily use over a long period.

13. Charcoal which is used in decolouring raw sugar is :

(a) Wood charcoal (b) Sugar charcoal
(c) Animal charcoal (d) Coconut charcoal

U.P.P.C.S. (Pre) 1998

Ans. (c)

Animal charcoal also, known as bone charcoal is primarily used for filtration and decolourization. Bone charcoal is often used in sugar refining as a decolourizing and de-ashing agent.

14. The main component of honey is :

(a) Glucose (b) Sucrose
(c) Maltose (d) Fructose

U.P.P.C.S. (Pre) 2002
I.A.S. (Pre) 1997

Ans. (d)

The main components of honey are fructose - 38.2%, glucose - 31.3%, sucrose - 1.3%, maltose - 7.1% and water - 17.2%. Fructose or fruit sugar is a simple ketonic monosaccharide found in many plants.

15. The sweetest sugar among the following is:

(a) fructose (b) glucose
(c) maltose (d) sucrose
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Fructose is the sweetest sugar among all natural sugars. Sugars are saccharides which have varying degrees of sweetness on a relative scale as illustrated in the following table :

Relative Sweetness Scale (Sucrose = 100)

Compound	**Sweetness**
Sucrose	100
Fructose	140-170
Glucose	70-80
Maltose	30-50
Galactose	35
Lactose	20

16. Given below are two statements, one is labelled as Assertion (A) and other as Reason (R) :

Assertion (A) : Invert sugar is more sweet than sucrose.

Reason (R) : Invert sugar is obtained by the hydrolysis of sucrose.

Select the correct answer from the code given below:

Code :

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

U.P. P.C.S. (Pre) 2020

Ans. (b)

Invert sugar is made by splitting disaccharide sucrose into its two components monosaccharides fructose and glucose (dextrose). The conventional way to make invert sugar is by the hydrolysis of sucrose to glucose and fructose. Invert sugar is sweeter than sucrose (ordinary white sugar) due to free fructose crystals in it which is the sweetest natural sugar. Hence, both (A) and (R) are true, but (R) is not the correct explanation of (A).

17. Match column I with column II and choose the correct answer using the code given below :

Column - I	**Column - II**
A. Grape Sugar	**(i) Fructose**
B. Fruit Sugar	**(ii) Sucrose**
C. Milk Sugar	**(iii) Natural Sweetner**
D. Table Sugar	**(iv) Glucose**
E. Stevia	**(v) Lactose**

Code :

	A	B	C	D	E
(a)	(ii)	(iv)	(v)	(iii)	(i)
(b)	(i)	(iii)	(iv)	(v)	(ii)
(c)	(iv)	(i)	(v)	(ii)	(iii)
(d)	(i)	(iv)	(v)	(iii)	(ii)

R.A.S./R.T.S (Pre) 2018

Ans. (c)

Grape Sugar	–	Glucose
Fruit Sugar	–	Fructose
Table Sugar	–	Sucrose
Stevia	–	Natural Sweetner
Milk Sugar	–	Lactose

18. Which of the following pairs is/are correctly matched?

1. Beet	**–**	**Sugar**
2. Honey	**–**	**Glucose and Fructose**
3. Cotton	**–**	**Cellulose**
4. Milk	**–**	**Lactose**

Select the correct answer using the code given below :

Code :

(a) Only 1, 2 and 3 (b) Only 2, 3 and 4
(c) Only 1, 2 and 4 (d) 1, 2, 3 and 4

U.P. R.O./A.R.O. (Pre) 2021

Ans. (d)

All of the given pairs are correctly matched. Hence, option (d) is the correct answer.

19. **Aspartame is an artificial sweetener sold in the market. It consists of amino acids and provides calories like other amino acids. Yet, it is used as a low-calorie sweetening agent in food items. What is the basis of this use?**
(a) Aspartame is as sweet as table sugar, but unlike table sugar, it is not readily oxidized in the human body due to lack of requisite enzymes.
(b) When aspartame is used in food processing, the sweet taste remains, but it becomes resistant to oxidation.
(c) Aspartame is as sweet as sugar but after ingestion into the body, it is converted into metabolites that yield no calories.
(d) Aspartame is several times sweeter than table sugar, hence food items made with small quantities of aspartame yield fewer calories on oxidation.

I.A.S. (Pre) 2011

Ans. (d)

Aspartame is an artificial, non-saccharide sweetener used as a sugar substitute in some foods and beverages. Aspartame is approx. 200 times sweeter than sucrose (table sugar). Due to this property, even though aspartame produces four kilo calories of energy per gram when metabolized, but the quantity of aspartame needed to produce a sweet taste is so small that its calorie contribution is negligible.

20. **Which of the following artificial sweeteners have the highest sweetness value in comparison to sucrose?**
(a) Aspartame (b) Alitame
(c) Sucrolose (d) Saccharin

Rajasthan P.C.S. (Pre) 2024

Ans. (b)

Relative sweetness to sucrose (table sugar) by weight of given artificial sweeteners is as follows :

Aspartame	–	200 times
Alitame	–	2000 times
Sucralose	–	600 times
Saccharin	–	200-700 times

Hence, option (b) is the correct answer.

21. **What is the basic of most useful classification of medications in medical chemistry?**
(a) Pharmacological effect (b) Molecular targets
(c) Chemical structure (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

Drugs usually interact with biomolecules such as carbohydrates, lipids, proteins and nucleic acids. These are called target molecules or drug targets. Drugs possessing some common structural features may have the same mechanism of action on targets. The classification based on molecular targets is the most useful classification in medical chemistry.

22. **Aspirin is obtained from –**
(a) Petroleum (b) Earth
(c) A tree (d) Chemical reaction of acids

47th B.P.S.C. (Pre) 2005

Ans. (c)

Aspirin is a salicylate. It works by reducing substances in the body that causes pain, fever and inflammation. It is sometimes used to treat or prevent heart attacks, strokes and chest pain. It is obtained from latex tree.

23. **Aspirin is –**
(a) Antibiotic (b) Antipyretic
(c) Reliever (d) None of the above

40th B.P.S.C. (Pre) 1995

Ans. (b)

Antipyretics cause the hypothalamus to override a prostaglandin-induced increase in temperature. The body then works to lower the temperature, resulting in a reduction in fever. Aspirin is antipyretic and analgesic.

24. **Antibiotics are mostly obtained from :**
(a) Fungi (b) Viruses
(c) Bacteria (d) Angiosperms

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

Antibiotics are medicines that fight bacterial infections in people and animals. They work by killing the bacteria or by making it hard for the bacteria to grow and multiply. Antibiotics are mostly obtained from bacteria. Majority of antibiotics are produced by Streptomyces (and other actinomycetes) and various Bacillus species. Actinomycetes is the family of soil bacteria that has produced most of our antibiotics and other medically useful molecules. Many antibiotics are also obtained from other microorganisms, such as moulds and fungus. With advances in medicinal chemistry, most modern antibiotics are semisynthetic modifications of various natural compounds. However, some are also produced solely by chemical synthesis in the lab.

25. **The antibiotic among the following is :**
(a) Pencillin (b) Aspirin
(c) Paracetamol (d) Sulfadiazine
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Among the given options, pencillin and sulfadiazine both are antibiotics. Aspirin and paracetamol are analgesic and antipyretic medicines.

26. **An example of antibiotic medicine :**
(a) Aspirin (b) Paracetamol
(c) Chloroquine (d) Penicillin

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

Among the given options, penicillin is an antibiotic medicine. Aspirin and paracetamol are analgesic and antipyretic medicines while chloroquine belongs to antimalarial medicines.

27. Which one of the following is not correctly matched?

(a) Antipyretic – Paracetamol
(b) Antifoaming – Polyamides Silicones
(c) Antiseptic – Aspirin
(d) Antirachitic – Calciferol (Vitamin D)

U.P.P.C.S. (Mains) 2006

Ans. (c)

Aspirin or acetylsalicylic acid is a salicylate drug and is generally used as an analgesic (something that relieves pain without producing anaesthesia or loss of consciousness) for minor aches and pains, to reduce fever (an antipyretic) and also as an anti-inflammatory drug. It is not antiseptic. Remaining pairs are correctly matched.

28. Which one of the following pairs is not correctly matched?

(a) Chloromycetin – antityphoid
(b) Crystal violet – antiseptic
(c) Quinine – antimalarial
(d) Aspirin – anaesthetic

UP. R.O./A.R.O. (Pre) 2017

Ans. (d)

Aspirin is not anaesthetic. It is analgesic (pain reliever) and antipyretic (to remove fever). Other pairs are correctly matched.

29. Which one of the following compounds is used as a sedative?

(a) Potassium Bromide (b) Calcium Chloride
(c) Ethyl Alcohol (d) Phosphorus Trichloride

U.P.P.C.S (Pre) 2010

Ans. (a)

Potassium Bromide (KBr) is a salt, widely used as an anticonvulsant and a sedative in the late 19th and early 20th centuries. It is used as a veterinary drug, as an antiepileptic medication for dogs. It is an odourless, colourless crystals or white granular solid with a pungent bitter saline taste.

30. Which one of the following forms an irreversible complex with a hemoglobin of the blood?

(a) Carbon Dioxide
(b) Pure Nitrogen gas
(c) Carbon Monoxide
(d) A mixture of Carbon Dioxide and Helium

U.P.P.C.S. (Pre) 1996

Ans. (c)

Carbon monoxide (CO) is a deadly, colourless, odourless, poisonous gas. It is produced by the incomplete burning of various fuels including coal, wood, charcoal, oil, kerosene, propane and natural gas. Carboxyhemoglobin (COHb) is a stable complex of carbon monoxide that generates in red blood cells when carbon monoxide is inhaled.

31. Milk is homogenized by :

(a) Adding a little sodium carbonate
(b) Removing its fat
(c) Breaking down fat particles to the microscopic size with the help of centrifuge
(d) Boiling only

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

Homogenization breaks the fat into small sizes so it no longer separates allowing the sale of non-separating milk at any fat specification. The fat in the milk normally separates from the water and collects at the top. Thus the consistency and texture is homogenized. It is a purely physical process, nothing is added to the milk.

32. Which one of the following oil is an active component of oil of clove?

(a) Menthol (b) Eugenol
(c) Methanol (d) Benzaldehyde

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

I.A.S. (Pre) 1997

Ans. (b)

Eugenol is a colourless or light yellow aromatic oily liquid extracted from cloves having chemical formula $C_{10}H_{12}O_2$. It smells like cloves with spicy pungent taste. It is the main component of oil of clove.

33. Which one of the following fruits is most suitable for jelly making?

(a) Mango (b) Papaya
(c) Guava (d) Wood apple

U.P.P.C.S. (Main) 2013

Ans. (c)

The smaller acid fruits are more suitable for jelly making since they are usually high in pectin content and acid. Guavas have high calcium and phosphorus contents. High pectin contents make guava suitable for jelly making.

34. Organic food is supposed to be better for us because it

(a) Relies on chemicals to improve the flavour
(b) Is more expensive to buy